Physical Design

Existing tools

Weighted Pen

Weighted Pen helps reduce tremor and is a good option for patients with Parkinson's Disease or decreased coordination. Write more legibly. Oversized grip.

Weighted pen has five 10 gram weights (included with the pen) that can be added or removed as needed. Weighted pen uses standard ballpoint pen refills.

Ergowriter

Thumb rest and finger cushions allow comfortable writing with either hand.

- The Ergowriter comfortably positions the thumb so that stress on the wrist is reduced. Designed to help relieve the pain of writing associated with osteoarthritis and other hand conditions.
- The Ergowriter can be used with either the left or right hand.
- Accommodates most standard pens and pencils.
- Blue ballpoint pen and soft writing grips are included.

Motor Movements: Using Insulin pen

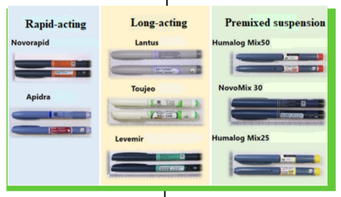
Before Injecting

- Select Needle Type → twist off cap
- Mixed types of Insulin: roll pen between palms & gently turn syringe up & down
- Installing a new pen needle
- tear protective sheet off → align needle head & turn into pen needle until tightly sealed

During Injection

- Pinch injection site with thumb & index finger
- Insert needle at 90° (vertical) until needle is completely inserted into subcutaneous tissue
- Press injection button all the way down

Current Insulin Pen / Injector Designs



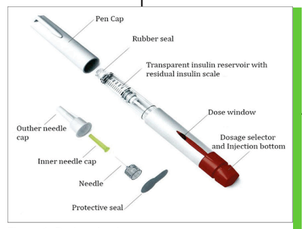
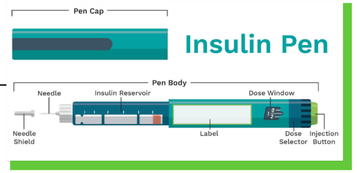
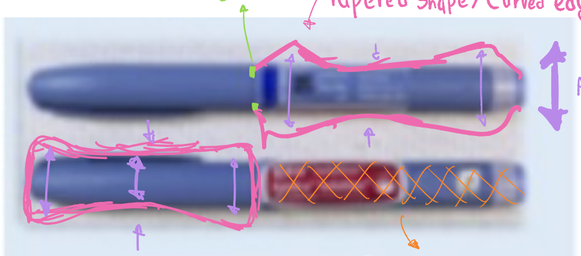


Figure 1: Design of syringe - pen





Wider cap

instead of sleek plastic

↓

textured silicone